

SYNTHESIZER AGENT

Imagine you are a user who has been wearing an AR/VR headset for an extended period, during which the device continuously recorded your surroundings.

TASK DESCRIPTION

From this full recording, I will select a short clip: video V. Your job is to envision a daily-life scenario S that occurs any amount of time after the events shown in video V have ended. Within this scenario S, think of a question Q that the user might naturally ask the AR/VR device; the answer to this question must require the device to review video V. The question Q should be rooted in everyday life, described as unambiguously as possible, and fully consistent with scenario S. Then create a multiple-choice question with 5 options where one is the correct answer.

CONTEXT YOU WILL RECEIVE

1. ****Segment list****: an ordered set of action descriptions in video V with timestamp.
2. ****QA key frame****: a single frame image from video V with timestamp, showing visual content at that moment.
3. ****Selected Object****: One object from the key frame has been specifically chosen, with its bounding box coordinates. Your eventual question or answer ****MUST**** be related to this selected object.

TOOLS YOU CAN CALL

You ****CANNOT**** see raw video, but you can use tools to gather more information:

****GET_SEGMENT(start_second, end_second)****

****GET_FRAME(timestamp_second)****

##Complete overall task in following steps:

1. Analyze full segment list and the QA key-frame image. Identify the selected object information provided in the context.
2. Brainstorm situations in which you might ask the device a daily-life question that requires information from the video V to answer.
3. Use the GET_SEGMENT and GET_FRAME tools to gather additional information needed for giving the question and answer.
 - After each function response, briefly reflect on what you learned before deciding whether another call is necessary.
 - Feel free to chain function calls: study responses, think, then request another refinement until you believe you understand enough to craft a good question-and-answer pair.
4. Use the given tools to cross-verify each facts in QA.
 - ****CRITICAL**** Questions and Answers must be supported by facts from at least ****TWO**** independent sources (frames or segment analyses)
5. Generate 5 multiple choice options (A, B, C, D, E) where one is correct and four are challenging but incorrect distractors.
 - The distractors should be plausible and close in meaning to confuse someone who hasn't paid close attention
 - All options should be stylistically consistent in length, structure, and detail level
6. When satisfied with your MCQ, call REQUEST_REVIEW to submit for review.
 - Include the question, all 5 options, correct answer letter, and evidence timestamps
 - The reviewer will provide feedback or approve with "pass"
 - If feedback is provided, refine your MCQ based on the feedback and submit for review again

Final output format

Use REQUEST_REVIEW tool with the following parameters:

- question: The question you generated
- option_a through option_e: The five answer options
- correct_answer: The letter of the correct answer (A/B/C/D/E)
- evidence_timestamps: Array of timestamp numbers that support your answer